A Technical Introduction to Logistic regression

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$P(y=1 \mid x) = \sigma(w^T x + b)$$

Section 1

The regression coefficients β0, β1, ..., βm are grouped into a single vector β of size m + 1. For each data point i, an additional explanatory pseudo-variable x0,i is added, with a fixed value of 1, corresponding to the intercept coefficient β0. The resulting explanatory variables x0,i, x1,i, ..., xm,i are then grouped into a single vector Xi of size m + 1. This makes it possible to write the linear predictor function as follows:

Section 2

■ ^ ϕ = K (y ¯ ln ■ (y ¯) + (1 − y ¯) ln ■ (1 − y ¯)) {\displaystyle {\hat {\ell }}_{\varphi }=K(\,{\overline {y}}\ln({\overline {y}})+(1-{\overline {y}})\ln(1-{\overline {y}}))}

Section 3

which shows that this formulation is indeed equivalent to the previous formulation. (As in the two-way latent variable formulation, any settings where β = β 1 − β 0 {\displaystyle {\boldsymbol {\beta }}={\boldsymbol {\beta }}_{1}-{\boldsymbol {\beta }}_{0}} will produce equivalent results.) Most treatments of the multinomial logit model start out either by extending the "log-linear" formulation presented here or the two-way latent variable formulation presented above, since both clearly show the way that the model could be extended to multi-way outcomes. In general, the presentation with latent variables is more common in econometrics and political science, where discrete choice models and utility theory reign, while the "log-linear" formulation here is more common in computer science, e.g. machine learning and natural language processing.

Section 4

β 0 = − 3 {\displaystyle \beta _{0}=-3} is the y-intercept. It is the log-odds of the event that y = 1 {\displaystyle y=1} , when the predictors x 1 = x 2 = 0 {\displaystyle x_{1}=x_{2}=0} . By exponentiating, we can see that when x 1 = x 2 = 0 {\displaystyle x_{1}=x_{2}=0} the odds of the event that y = 1 {\displaystyle y=1} are 1-to-1000, or 10 − 3 {\displaystyle 10^{-3}} . Similarly, the probability of the event that y = 1 {\displaystyle y=1} when x 1 = x 2 = 0 {\displaystyle x_{1}=x_{2}=0} can be computed as 1 / (1000 + 1) = 1 / 1001. {\displaystyle 1/(1000+1)=1/1001.}

Section 5

Y i 0 ∗ = β 0 ⋅ X i + ε 0 Y i 1 ∗ = β 1 ⋅ X i + ε 1 {\displaystyle {\begin{aligned}Y_{i}^{0\ast }&={\boldsymbol {\beta }}_{0}\cdot \mathbf {X} _{i}+\varepsilon _{0}\,\\Y_{i}^{1\ast }&={\boldsymbol {\beta }}_{1}\cdot \mathbf {X} _{i}+\varepsilon _{1}\,\end{aligned}}}

Section 6

ε 0 ∼ EV 1 ■ (0 , 1) ε 1 ∼ EV 1 ■ (0 , 1) {\displaystyle {\begin{aligned}\varepsilon _{0}&\sim \operatorname {EV} _{1}(0,1)\\\varepsilon _{1}&\sim \operatorname {EV} _{1}(0,1)\end{aligned}}}

Section 7

where μ is a location parameter (the midpoint of the curve, where p (μ) = 1 / 2 {\displaystyle p(\mu)=1/2}) and s is a scale parameter. This expression may be rewritten as:

Section 8

Predictions The ■ β 0 {\displaystyle \beta _{0}} ■ and ■ β 1 {\displaystyle \beta _{1}} ■ coefficients may be entered into the logistic regression equation to estimate the probability of passing the exam. For example, for a student who studies 2 hours, entering the value x = 2 {\displaystyle x=2} into the equation gives the estimated probability of passing the exam of 0.25:

Section 9

■ k = { − ln ■ p k if y k = 1 , − ln ■ (1 − p k) if y k = 0. {\displaystyle \ell _{k}={\begin{cases}-\ln p_{k}&{\text{ if }}y_{k}=1,\\-\ln(1-p_{k})&{\text{ if }}y_{k}=0.\end{cases}}}

Section 10

and since Y ∈ { 0 , 1 } {\displaystyle Y\in \{0,1\}} , we see that Pr (y ■ X ; θ) {\displaystyle \Pr(y\mid X;\theta)} is given by Pr (y ■ X ; θ) = h θ (X) y (1 − h θ (X)) (1 − y) . {\displaystyle \Pr(y\mid X;\theta)=h_{\theta }(X)^{y}(1-h_{\theta }(X))^{(1-y)}.} We now calculate the likelihood function assuming that all the observations in the sample are independently Bernoulli distributed,

Section 11

Pr (Y i = 0) = 1 Z e β 0 ⋅ X i Pr (Y i = 1) = 1 Z e β 1 ⋅ X i {\displaystyle {\begin{aligned}\Pr(Y_{i}=0)&={\frac {1}{Z}}e^{{\boldsymbol {\beta }}_{0}\cdot \mathbf {X} _{i}}\\[5pt]\Pr(Y_{i}=1)&={\frac {1}{Z}}e^{{\boldsymbol {\beta }}_{1}\cdot \mathbf {X} _{i}}\end{aligned}}}

Section 12

logit ■ (E ■ [Y i n i | X i]) = logit ■ (p i) = ln ■ (p i 1 − p i) = β ⋅ X i , {\displaystyle \operatorname {logit}

\left(\operatorname {\mathbb {E} } \left[\left.{\frac {Y_{i}}{n_{i}}}\,\right|\,\mathbf {X} _{i}\right]\right)=\operatorname {logit} (p_{i})=\ln \left({\frac {p_{i}}{1-p_{i}}}\right)={\boldsymbol {\beta }}\cdot \mathbf {X} _{i}\,,}

Section 13

where H (Y ■ X) {\displaystyle H(Y\mid X)} is the conditional entropy and D KL {\displaystyle D_{\text{KL}}} is the Kullback–Leibler divergence. This leads to the intuition that by maximizing the log-likelihood of a model, you are minimizing the KL divergence of your model from the maximal entropy distribution. Intuitively searching for the model that makes the fewest assumptions in its parameters.

Section 14

Deviance and likelihood ratio test ■ a simple case In any fitting procedure, the addition of another fitting parameter to a model (e.g. the beta parameters in a logistic regression model) will almost always improve the ability of the model to predict the measured outcomes. This will be true even if the additional term has no predictive value, since the model will simply be "overfitting" to the noise in the data. The question arises as to whether the improvement gained by the addition of another fitting parameter is significant enough to recommend the inclusion of the additional term, or whether the improvement is simply that which may be expected from overfitting. In short, for logistic regression, a statistic known as the deviance is defined which is a measure of the error between the logistic model fit and the outcome data. In the limit of a large number of data points, the deviance is chi-squared distributed, which allows a chi-squared test to be implemented in order to determine the significance of the explanatory variables. Linear regression and logistic regression have many similarities. For example, in simple linear regression, a set of K data points (xk, yk) are fitted to a proposed model function of the form y = b 0 + b 1 x {\displaystyle y=b_{0}+b_{1}x} . The fit is obtained by choosing the b parameters which minimize the sum of the squares of the residuals (the squared error term) for each data point: